FitTech Data Science Capstone

Assessing App Usage & Calorie Burn Patterns to Drive Personalized Engagement

Project Objectives

Understanding the modern fitness user to optimize platform experiences and maximize retention.

Identify Demographics

Analyze core user traits and subscription behaviours.

Workout Preferences

Map exercise choices to calorie burn efficiency.

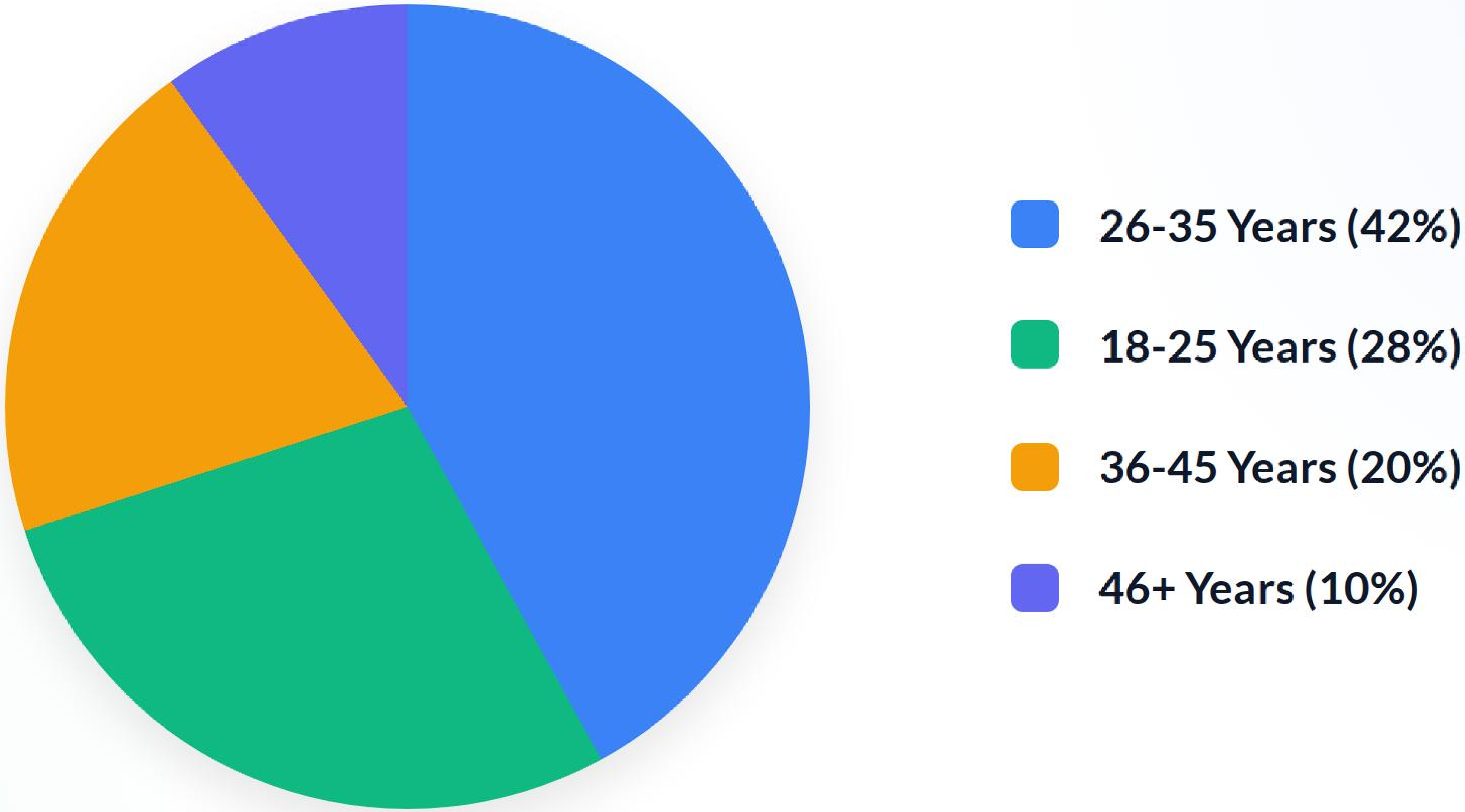
UI Engagement

Evaluate notification triggers and completion rates.

Predictive ML

Build classification models to predict highly engaged users.

User Demographics



The 26-35 age bracket forms the foundation of our user base, indicating a strong market fit among young professionals.

Workout Preferences & Engagement

Users strongly prefer "Mixed" routines over single-discipline workouts, demanding a varied content library.

Activity & Calorie Burn Analysis

Workout Type	Average Duration	Avg Calories Burned	Total Avg Steps
Mix	55 mins	450 kcal	5,200
Cardio	45 mins	420 kcal	6,500
Strength	50 mins	380 kcal	2,100
Yoga	40 mins	210 kcal	800

*Data extracted from **FitTech_Final_Dataset.xlsx**. Mixed workouts yield the highest calorie burn and longest session durations.*

App Engagement Metrics

82%

Workout Completion Rate

75%

Notification Click-Through

Healthy UI Interaction

Our UI engagement is incredibly healthy. A 75% click-through rate on push notifications indicates that users are highly receptive to our prompts.

This receptiveness is a vital lever for our machine learning strategy, allowing us to deploy personalized reminders with high confidence in conversion.

Key Engagement Drivers

Smart Reminders

Timely push notifications significantly increase daily active use and prevent user churn in the critical first 30 days.

Goal Alignment

Users who input specific targets like "Weight Loss" during onboarding show a 2x higher long-term retention rate.

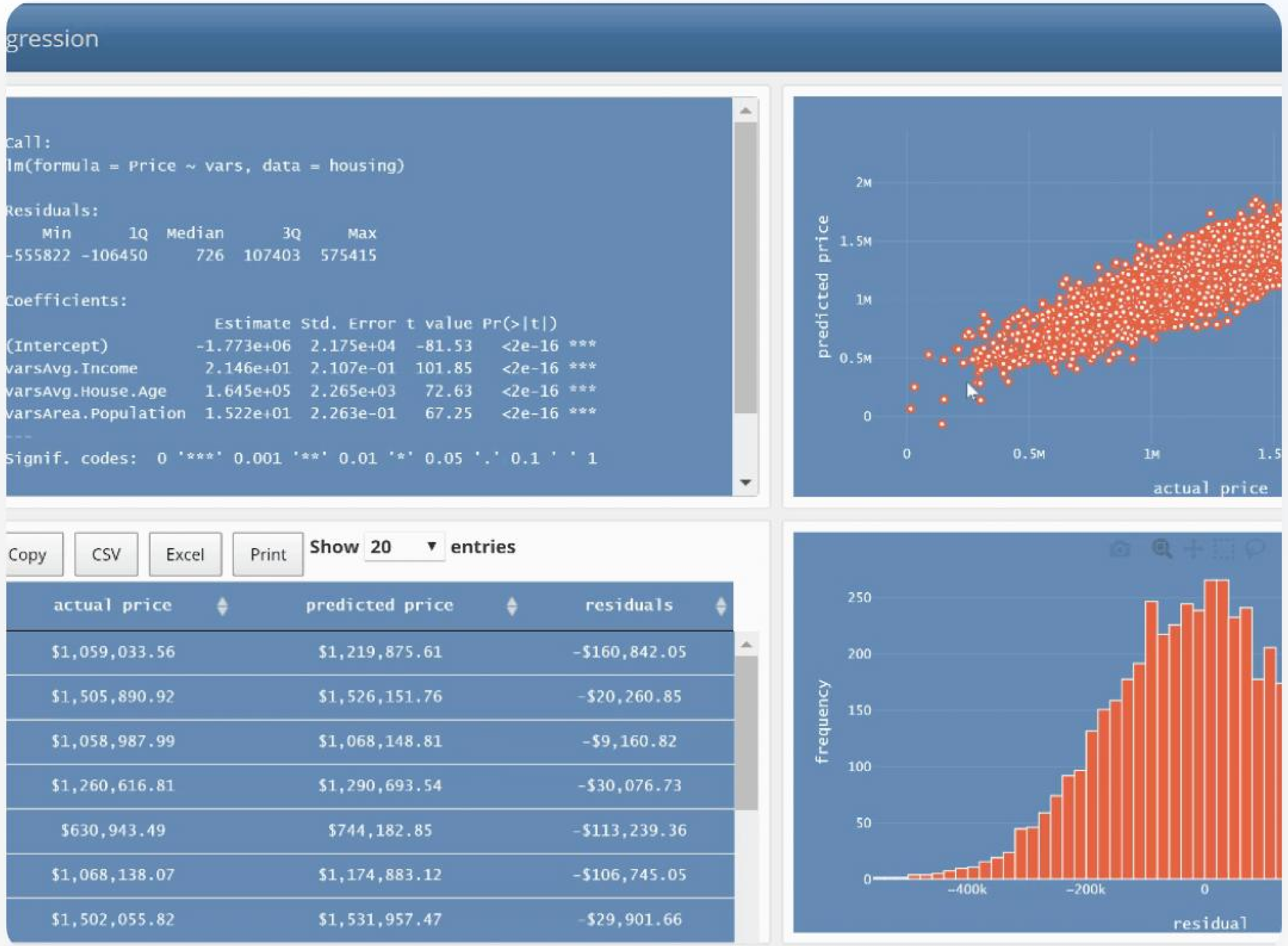
UI Friction

Simplified, one-tap workout logging directly correlates to higher session completion and accurate calorie tracking.

Predictive Modeling (ML)

We deployed several classification models to predict highly engaged users, relying on features standardized via StandardScaler and OneHotEncoder.

- **Classification Models:** Logistic Regression, KNN, and Decision Trees were evaluated.
- **Dataset Outcomes:** Detailed results are documented in **ML_Model_Results.xlsx**.
- Logistic Regression performed exceptionally well at flagging users likely to convert to paid subscriptions.



Logistic Regression Probability:

$$P(Y=1) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\dots)}}$$

Forecasting Session Time (Regression)

Linear Regression forecast: If we retain a user through the first 4 weeks, their session duration steadily climbs and plateaus at a highly profitable engagement level.

Aligning with 2026 Trends

AI Personalization

Deploying adaptive ML algorithms to generate highly personalized, evolving workout routines.

Wearable IoT

Deeper, real-time synchronization with smartwatches and heart rate monitors.

VR Fitness

Integrating immersive virtual reality and augmented reality guided sessions.

Mental Wellness

Expanding tracking to include mindfulness, sleep quality, and holistic wellness.

Strategic Recommendations

Deploy ML-Driven Personalization

Transition from static workout plans to AI-curated routines based on the Logistic Regression engagement probability scores.

Optimize Push Notifications

Capitalize on the 75% click-through rate by targeting users during their historically active windows rather than generic broadcast times.

Market 'Mix' Routines

Target the dominant 26-35 demographic with premium 'Mix' workout subscription tiers, as these yield the highest calorie burn and session duration.

Invest in Backend Syncing

Enhance database architecture to support frictionless, real-time wearable data sync, reducing user logging fatigue.

Questions?

Thank you for your time and attention.

Image Sources

https://cdn.dribbble.com/userupload/43795567/file/original-51ed717d23efa0180777cee011f33e78.png?resize=752x&vertical=center

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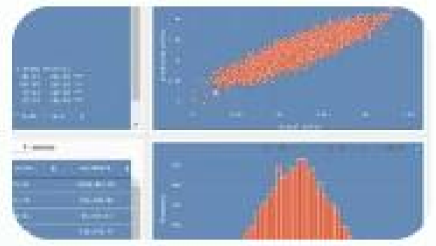

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